

Prompt-CAST: Text-Conditioned Image Inpainting with a Frozen Vision-Language Prior

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Abstract—We present Prompt-CAST, a text-conditioned extension to the CAST-Net image-inpainting architecture. A prompt embedding from a frozen CLIP text encoder modulates the Cross-Scale Affinity Mixer's hole queries via a FiLM layer before the affinity computation, so a short text prompt can steer what a masked region is filled with. The design is additive by construction: a zero-initialized FiLM head, or the absence of a prompt, reduces the module to CAST-Net's original unconditional behaviour exactly. This paper's contribution at this stage is the architecture and a fully verified, reproducible implementation — the FiLM-conditioned module, a frozen CLIP text-encoder wrapper, a manifest pipeline that sources ADE20K prompts from the dataset's own scene-category labels with no separate captioning model, and training-script support for the conditioned path with CUDA/MPS/CPU device selection — covered by a test suite that checks real behaviour (gradient flow into the FiLM head, exact known-pixel preservation under conditioning, and a real, if small-scale, training run on genuine ADE20K images with real CLIP-derived prompts showing consistent loss convergence). No claim is made about inpainting quality, prompt adherence, or efficiency relative to diffusion-based text-guided inpainting: full-scale training, a CLIP-score evaluation, and baseline comparisons have not yet been run, and this paper reports that honestly as the immediate next step rather than presenting unverified numbers.

Keywords—text-guided image inpainting; FiLM conditioning; frozen vision-language prior; CLIP; continual reproducibility; sparse attention

1. Introduction

Text-to-image and text-guided inpainting systems built on diffusion models have demonstrated strong controllability: a short natural-language prompt can specify what should appear inside a masked region, not merely that the region should look plausible given its surroundings. This controllability comes at a cost. Diffusion-based inpainting ordinarily requires iterative sampling, and text conditioning is usually threaded through a large, jointly trained denoising network, making these systems expensive to run and difficult to deploy where latency matters.

CAST-Net is a compact, single-pass inpainting architecture built around three components: Sparse Global Context (SGC), a Cross-Scale Affinity Mixer (CSAM), and Frequency-Aware

Residual Refinement (FARR). CSAM already performs a form of content selection: it retrieves texture for the masked region by computing affinities between queries sampled inside the hole and keys/values drawn from observed source locations. This paper asks whether that same retrieval mechanism can be steered by a text prompt — without discarding CAST-Net's efficiency advantage by attaching a heavy conditioning pathway.

Prompt-CAST answers this with the smallest change that could plausibly work: a FiLM (Feature-wise Linear Modulation) layer that reshapes CSAM's hole queries using an embedding from a frozen CLIP text encoder, before the existing cosine-similarity affinity computation. The text encoder is never fine-tuned; only a small per-channel scale-and-shift head is trained. Because the modulation is additive around the identity operation, the unconditional CAST-Net model is recovered exactly whenever no prompt is supplied.

The contributions of this paper are: (1) a FiLM-conditioned extension of CSAM that adds text control to CAST-Net's texture retrieval with no architectural change when unused; (2) a manifest pipeline that sources ADE20K text prompts from the dataset's own scene-category labels, requiring no separate captioning model or additional annotation; (3) a fully tested, reproducible reference implementation, including training-script support for the conditioned path and automatic CUDA/MPS/CPU device selection; and (4) an explicit, honest account of what has and has not been evaluated, since the central empirical question — whether text conditioning measurably improves controllability without harming reconstruction quality — has not yet been tested at a scale that would support a claim either way.

2. Related Work

CAST-Net [1] is the base architecture this work extends; its SGC, CSAM, and FARR components, and their single-pass efficiency profile, are unchanged here except for the FiLM addition to CSAM described in Section 3. CLIP [2] provides the frozen vision-language embedding used for text conditioning; its text tower is used purely as a fixed feature extractor, consistent with how CLIP embeddings are commonly reused for conditioning generative models without fine-tuning the encoder itself. FiLM [3] is the general conditioning mechanism adapted here: a learned affine transform applied to intermediate features, conditioned on an external embedding. Large-scale text-guided diffusion inpainting systems, including latent diffusion approaches [4], represent the higher-capacity, higher-latency alternative this design is positioned against; Section 6 lists the comparison against such a system as unevaluated future work rather than a claim made in this paper.

3. Method

3.1 FiLM-Conditioned Cross-Scale Affinity Mixer

CAST-Net's CSAM computes a cosine-similarity affinity between a hole query q_i and keys k_j drawn from observed source locations Ω_v , normalized by a softmax over Ω_v with temperature τ , and retrieves texture as a weighted sum of the corresponding values v_j . Prompt-CAST modulates

the query before this computation. Given a frozen text embedding e derived from the prompt, a small linear head produces a per-channel scale and shift:

$$(\gamma, \beta) = \phi(e), \gamma, \beta \in \mathbb{R}^d$$

The query is modulated and re-normalized before the affinity computation, so the modulation actually changes which source locations are attended to, rather than being renormalized away:

$$\tilde{q}_i = (I + \gamma) \odot q_i + \beta, \quad \hat{q}_i = \tilde{q}_i / \|\tilde{q}_i\|_2$$

The affinity computation, texture retrieval, and gating that follow are otherwise unchanged from CAST-Net's original CSAM. ϕ is initialized to output exactly zero (both weight and bias), so at initialization, and whenever no prompt is supplied, $\hat{q}_i = q_i$ and the module is bit-for-bit identical to unconditional CSAM. This identity property is not asserted; it is tested directly (Section 4).

3.2 Frozen Text Encoder

Prompts are encoded with a frozen CLIP text tower (ViT-B/32, OpenAI weights, loaded with the matching QuickGELU activation configuration). No component of the text encoder is updated during training; only ϕ is trainable. This keeps the parameter cost of text conditioning to a small linear head rather than a second large network.

3.3 Text-Prompt Supervision Without a Captioning Model

Training a text-conditioned model requires (image, mask, prompt) triples. Rather than generating captions with a separate model, Prompt-CAST sources prompts directly from ADE20K's own scene-category annotations (e.g. the category “airport_terminal” becomes the prompt “airport terminal”), via a manifest-construction script that pairs each image with its category and a content hash. This is weaker supervision than a region-specific caption — the prompt describes the whole scene, not necessarily the masked region specifically — but it requires no additional model or annotation effort and is available for the full dataset at no extra cost.

4. Verified Implementation and Reproducibility

Every component described in Section 3 is implemented and covered by tests that check actual behaviour rather than interface shape alone. Specifically: (a) a zero-initialized FiLM head reproduces the unconditional model's output exactly, verified within a single model instance to avoid the confound of comparing separately constructed models with different parameter counts, whose random initialization would otherwise diverge even under an identical seed; (b) gradients reach the FiLM head during backpropagation; (c) the compositing guarantee that observed pixels are preserved exactly in the output holds under text conditioning, not only in the unconditional path; and (d) a synthetic training-loop smoke test confirms that one full optimization step — forward pass, loss, backward pass, optimizer update, checkpoint save and reload — behaves correctly through the conditioned path.

Beyond synthetic checks, the manifest pipeline, the frozen text encoder, and the training script were run against genuine ADE20K data. A manifest built from real ADE20K images and the dataset's own sceneCategories.txt file, real CLIP embeddings computed from the resulting

prompts, and a real training run (24 images, 16 base channels, 128×128 resolution, 15 epochs, Apple MPS acceleration) produced a generator loss that decreased consistently across training — from approximately 0.45–0.56 at the start to 0.24–0.32 by the final epoch — confirming the full pipeline is mechanically correct and trainable, not merely free of runtime errors. This run used a deliberately reduced model capacity and dataset size and is not offered as an evaluation result; its only purpose is confirming the training mechanics work correctly before committing to full-scale training.

The reference implementation, including the test suite referenced above, is publicly available at <https://github.com/maheshchudaman/Prompt-CAST>. Training-script support additionally includes automatic device selection across CUDA, Apple's MPS backend, and CPU, with the selected device recorded in each run's provenance metadata.

5. Preliminary Reference Point

This paper reports one verified, real result: the unconditional CAST-Net-full baseline on ADE20K, from a seed-42 training run archived with full per-image evidence (2,021 test images, checkpoint SHA-256 723b61c5..., archived at <https://github.com/maheshchudaman/CAST-Net-Full-Seed42-RTX4050>). This is not a Prompt-CAST result: it contains no text conditioning and is reported here solely as the reference point that a trained text-conditioned model will need to be compared against.

Hole PSNR: 19.87 ± 3.33 dB | Full-image SSIM: 0.900 ± 0.050 | FID: 10.47 | Full-image LPIPS: 0.070 ± 0.034

6. Limitations and Planned Evaluation

The following have not been done, and no claim in this paper should be read as implying otherwise:

- Full-scale text-conditioned training — the complete ADE20K training set, at CAST-Net's original resolution (256×256) and capacity (64 base channels), has not been trained with text conditioning active. The run described in Section 4 used a reduced model and 24 images specifically to validate mechanics, not quality.
- A CLIP-score (or equivalent prompt-adherence metric) has not been computed. There is currently no evidence, in either direction, that the completed regions reflect their prompts.
- No comparison exists yet between the text-conditioned model and the unconditional CAST-Net baseline reported in Section 5, so it is not known whether conditioning helps, hurts, or has no measurable effect on reconstruction quality.
- No comparison exists against a pretrained diffusion-based text-guided inpainting system, so the efficiency claim motivating this design (Section 1) has not been empirically tested.
- Prompt supervision is scene-level (Section 3.3), not region-specific; whether this is sufficient signal for the model to learn meaningful hole-specific control, as opposed to global style, is an open question this paper does not resolve.

These four gaps define the next stage of this work directly: full-scale training on GPU hardware, CLIP-score evaluation, and the two comparisons above are the concrete, measurable steps between the verified architecture reported here and a defensible empirical claim about text-conditioned inpainting.

7. Conclusion

This paper introduced Prompt-CAST, a text-conditioned extension to CAST-Net that steers the Cross-Scale Affinity Mixer's texture retrieval via a FiLM-modulated query, driven by a frozen CLIP text encoder. The design is intentionally minimal: the modulation reduces to the exact identity operation when no prompt is supplied or when the FiLM head is zero-initialized, so text conditioning is additive to CAST-Net's existing behaviour rather than a competing architecture, and the frozen text encoder adds no trainable parameters beyond a small per-channel scale-and-shift head.

Beyond the architecture itself, this work's primary contribution at this stage is a fully verified, reproducible implementation. Every component — the FiLM-conditioned CSAM module, the frozen CLIP text-encoder wrapper, the ADE20K prompt-manifest pipeline built from the dataset's own scene-category labels, and the training script's text-conditioning path — is covered by tests that exercise real behaviour, not just interface shape: gradient flow into the FiLM head, exact preservation of known pixels under conditioning, and end-to-end execution of a real, if small-scale, training run on genuine ADE20K images with real CLIP-derived prompts, which showed the generator loss decreasing consistently across 15 epochs. This confirms the method is mechanically sound and trainable; it does not yet establish that text conditioning improves — or even preserves — inpainting quality at scale, and it makes no claim about controllability, since no metric for prompt-adherence has been computed against a trained checkpoint.

What remains before this method can be evaluated as a contribution, rather than a design: full-scale training on the complete ADE20K training set at the resolution and capacity used for CAST-Net's own reported results, a CLIP-score measurement to establish whether the completed regions actually reflect their prompts, a comparison against the unconditional CAST-Net baseline to confirm conditioning does not degrade base quality, and a comparison against at least one pretrained diffusion-based text-guided inpainting method to test the efficiency claim this design is built around. None of these evaluations have been run, and no number in this direction should be treated as established until they are.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work.

Data and Code Availability

The Prompt-CAST reference implementation, including its test suite, is available at <https://github.com/maheshchudaman/Prompt-CAST>. The unconditional CAST-Net baseline

reported in Section 5, including per-image evidence and checkpoint hashes, is available at <https://github.com/maheshchudaman/CAST-Net-Full-Seed42-RTX4050>. ADE20K [5] must be obtained under its own terms; this paper redistributes no dataset files.

Use of AI-Assisted Tools

AI-assisted tools were used for language editing, document formatting, and reference checking, in accordance with the target journal's policy on AI-assisted writing. The authors verified the scientific content, code, figures, and final text and accept full responsibility for the manuscript.

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